



Tatini Pavitra Strotas

Tagline: Restoring Purity, Preserving Culture.

Welcome to the presentation for the CM YUVA Innovation Challenge. Tatini Pavitra Strotas our vision is to create a harmonious ecosystem where modern technology restores river purity while preserving the cultural sanctity of India's sacred ghats. We are starting our journey in Varanasi – the spiritual heart of India.

Presented by ATHARVA DWIVEDI to the esteemed judges and stakeholders from the Government of Uttar Pradesh, E-Cell IIT Kanpur.

The Problem: The "Efficiency Gap"

Despite significant investments, the sacred River Ganga at Varanasi continues to suffer from pollution. The core issue isn't a lack of funding, but rather an **inefficient last-mile execution and management** of existing solutions.

1

Investment vs. Outcome

Over ₹20,000 crore has been allocated under the *Namami Gange* project. However, only approximately 69% of these funds were utilized by 2024–25, and a mere 52% of the targeted sewage treatment capacity is operational.

2

The Sewage Crisis

India generates an alarming 72,000 MLD of sewage daily, yet only around 28% of it is effectively treated. Urban areas like Varanasi contribute disproportionately to this crisis, leading to significant river pollution.

3

Varanasi's Unique Challenge

Varanasi boasts over 80 ghats, which are used daily for religious rituals. During major festivals, the city experiences a **3–4x increase in solid waste**, exacerbating the pollution problem.



Cultural Friction in Existing Solutions

Many modern interventions aimed at cleaning the Ganga have unfortunately failed. This is primarily because they **ignore ritual practices, clash with deeply held faith**, and treat the ghats purely as infrastructure rather than the sacred spaces they are.

This oversight often creates a misconception: **“Culture and modernization can't go hand in hand.”**

The reality is, without **cultural alignment and understanding**, even the most well-funded and technologically advanced projects are destined to underperform. Our approach recognizes this critical gap.

The Solution: Bridging the Efficiency Gap

Core Philosophy

Our guiding principle is “**Modern Outlook without Cultural Harm.**” We believe in leveraging advanced solutions that respect and integrate with existing cultural practices.

The Tatini Approach

Tatini Pavitra Strotas is not just a business; it's a reform movement. We aim to create “**Pavitra Strotas**” — clean, functional sources of purity that blend seamlessly into the ghat culture, offering traditional aesthetics with silent, efficient technology.

Designed for Continuous Use

Our solutions are engineered for continuous operation, even during **peak ritual periods and festivals**, ensuring uninterrupted service and purity.

Strategic Framing: At this stage, **we are proposing a phased, data-validated pilot designed to deliver measurable outcomes** without cultural disruption, directly bridging the efficiency gap.



Product & Service Ecosystem: Smart Sanitation Infrastructure

Our ecosystem begins with advanced, yet culturally sensitive, smart sanitation infrastructure designed for the unique demands of sacred sites.

Smart Toilets & Changing Rooms

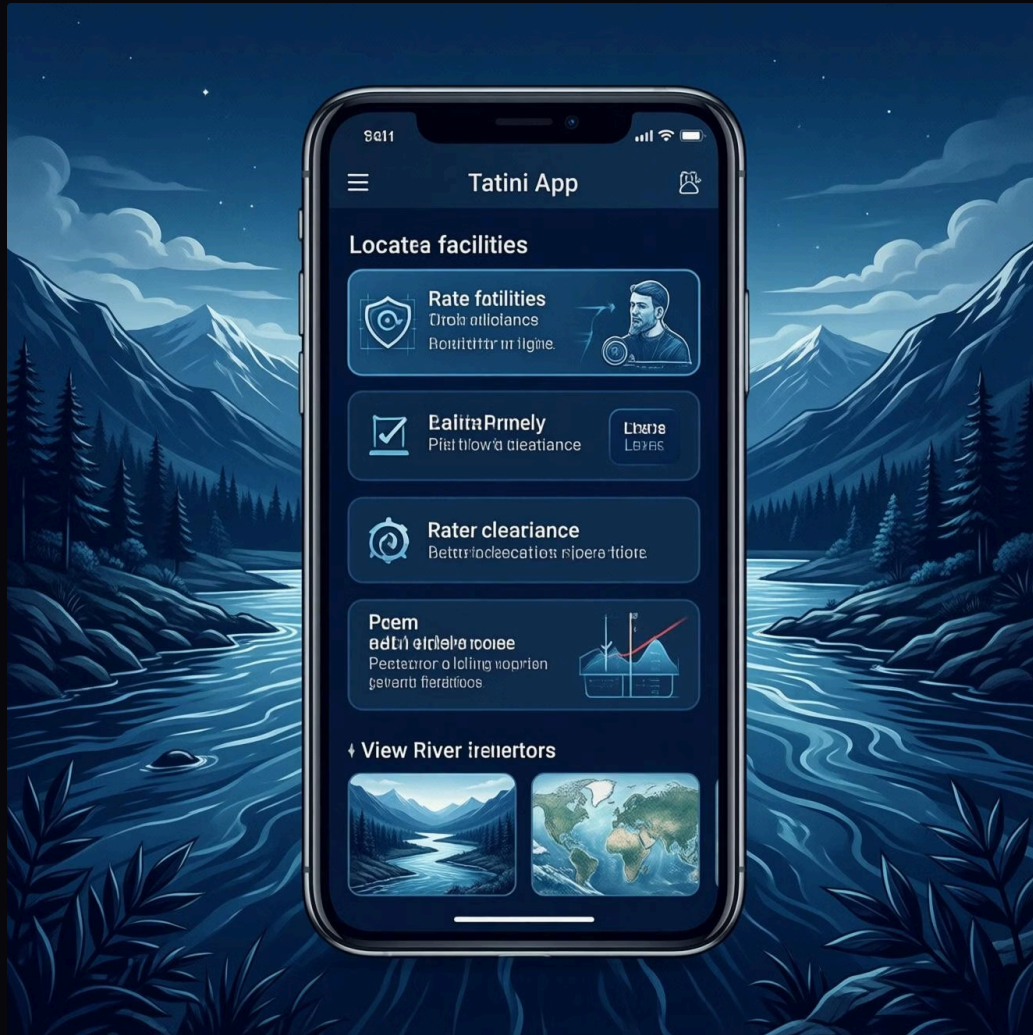
Inspired by the large-scale models proven effective at Sangam and Kumbh Mela, our facilities are designed for **high footfall and minimal downtime**.

Hygiene First

We implement automated cleaning cycles, touch-free systems, and rigorous protocols to ensure **>99% hygiene compliance** even in the busiest zones.

These modular units are not only functional but also designed to harmonize with the traditional aesthetics of the ghats, ensuring they are seen as an enhancement, not an intrusion.

Smart Feedback & Usage: The "Tatini App"



Pilgrim & Tourist Companion

- **Locate Nearest Clean Facility:** Instant access to information on nearby hygienic facilities.
- **Rate Cleanliness:** Real-time feedback mechanism allows users to report and rate the cleanliness of facilities, ensuring accountability.
- **View River Health Indicators:** Access to key environmental data such as Dissolved Oxygen (DO) levels and turbidity, fostering transparency and trust.

Revenue Layer

- **Subscription Services:** Offer premium utilities and enhanced feedback management.
- **Data-Driven Maintenance Prioritization:** Utilize user feedback to efficiently allocate resources and prioritize maintenance tasks.

IoT-Powered Monitoring & Maintenance

We employ advanced, contract-based IoT monitoring to ensure the continuous efficiency and proactive maintenance of our installations.



Real-time Sensor Tracking

Our deployed sensors continuously monitor critical parameters such as facility usage frequency, water levels, and operational status, providing immediate insights.



Automated Maintenance Alerts

The system automatically generates alerts for potential issues or maintenance requirements, ensuring timely intervention before minor problems escalate.



Preventing Festival Breakdowns

Proactive monitoring prevents critical breakdowns, especially during high-traffic festivals, guaranteeing uninterrupted service when it's needed most.



Reduced Operational Costs

By automating surveillance and alerts, we significantly reduce the need for constant manual monitoring, leading to substantial cost savings + **OUR IOT SENSORS EVEN CHEAPER THAN EXISTING USED BY GOVERNMENT AND ARE EFFICIENT AND TESTED BY SATYAM A PHD SCHOLAR FROM IIT GUWAHATI HONOURED FOR THE SAME BACK IN 2024.**

This intelligent system exemplifies our commitment to [efficiency through automation](#), rather than relying on manpower overload, ensuring consistent purity with minimal human intervention.



SUSTAINABILITY

Business & Revenue Model

Our robust business model is designed to ensure financial viability and scalable growth, leveraging both government and consumer channels.

B2G – Business to Government

- **Direct Government Contracts:** Securing contracts for comprehensive ghat utility management and sanitation O&M services.
- **Vendor Linkage & Task Tracking:** Implementing transparent systems for vendor management and date-wise task tracking, which prevents fund leakage and the underutilization of assets.

B2C – Business to Consumer

- **Pay-Per-Use / Subscription Facilities:** Offering accessible pay-per-use options or subscription models for our premium sanitation facilities.
- **Premium Toilets & Changing Rooms:** Providing enhanced, high-quality toilet and changing room experiences for pilgrims and tourists seeking superior comfort and hygiene.

Additional Revenue Streams & Partnerships

Beyond our core B2G and B2C models, Tatini Pavitra Strotas will diversify its revenue through strategic channels, enhancing our financial resilience and impact.

Tourism Revenue Boost

Cleaner and more pleasant ghats directly contribute to an increase in tourism, bringing economic benefits to the local community and creating new revenue opportunities.

Data Monetization

The valuable environmental and usage data collected can be monetized by offering it to research institutions for academic studies or urban planning bodies for strategic development.

Annual Maintenance Contracts (AMC)

Offering AMCs for the upkeep and continued optimal performance of our installed infrastructure, ensuring long-term operational excellence.

CSR & ESG Partnerships

Collaborating with corporate social responsibility (CSR) initiatives and environmental, social, and governance (ESG) funds provides additional funding and promotes sustainable development.

Operational Efficiency & Continuous Improvement

Our operational strategy is built on principles of lean construction, empowered local workforce, and relentless iteration to ensure maximum impact and sustainability.

01

Cost-Effective Construction

We utilize modular, prefabricated units that allow for **faster deployment** compared to traditional concrete structures. These designs also feature flood-resilient capabilities, essential for monsoon conditions.

02

Workforce Strategy

Our core operational team will comprise **Eco-Tech Leads** — local youth trained specifically in the maintenance and management of bio-toilets and sensor systems, fostering community involvement and employment.

03

Testing & Iteration

We are committed to continuous pilot testing and model upgrades based on real-world data. This includes rigorous evaluation during **seasonal stress, festival loads, and direct user feedback**, ensuring our solutions are always improving.

This iterative approach ensures that our models are constantly refined for better versions, providing the most effective and sustainable solutions for purity and culture.

“The Cultural Push”

Our go-to-market strategy centers on integrating our initiatives deeply within the cultural fabric of the community, fostering a sense of ownership and responsibility from within.

Community Integration

- **Priests as River Protectors:** Positioning local religious leaders as key advocates for river cleanliness and responsible facility usage.
- **Cultural Messaging:** Framing the use of **Tatini facilities as an act of devotion (seva) to Maa Ganga**, aligning with deeply held spiritual values.
- **Direct Economic Benefit:** Demonstrating how cleaner ghats directly **enhance the livelihoods of boatmen, vendors, and local residents through increased tourism** and improved public health.

Youth Mobilization

- **Campus Ambassadors & Internships:** Partnering with institutions like **IIT-BHU, BHU, and local colleges to engage students as campus ambassadors and offer valuable internships.**
- **Educational Outreach:** Conducting **interactive seminars in schools and corporate** environments to educate and inspire the next generation.

This strategy cultivates a profound sense of responsibility, rather than relying on enforcement, ensuring long-term, sustainable behavioral change.



Impact & Vision

Our commitment to purity extends beyond immediate sanitation, generating a ripple effect of positive change across social, environmental, and economic spheres.

SOCIAL IMPACT

Enhancing community well-being and spiritual practices.

- Reduced exposure to polluted water
- Lower risk of water-borne diseases
- Safer and more dignified ritual practices

ENVIRONMENTAL IMPACT

Restoring ecological balance and natural beauty to the riverine ecosystem.

- 60–70% reduction in visible ghat waste
- Improved water quality indicators:
 - Dissolved Oxygen (DO)
 - Turbidity
 - Organic load near ghats

ECONOMIC IMPACT

Driving local prosperity and sustainable development.

- Faster execution of cleaning projects
- Increased religious tourism revenue
- Creation of sustainable local green jobs

These interconnected benefits underscore our holistic approach to creating lasting value for communities and the environment.

समापन संदेश

अविरल बहे गंगा की धारा, निर्मल हो उसका हर स्रोत,

शुद्ध रहे यह पुण्य प्रवाह, तभी बचेगा इसका बोध।

~ गंगा की पवित्रता केवल आस्था नहीं, हमारी सामूहिक जिम्मेदारी है।

Tatini Pavitra Strotas

Restoring purity. Respecting culture. Enabling technology.
Delivering impact.